Power Series and Taylor Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Radius and interval of convergence**

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

Ratio test: $\left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| \to |x|$. Converges for $|x| < 1$: radius of convergence $R = 1$.

2 **[Foundational]** • **Target: Radius and interval of convergence**

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

Ratio test: $\left|\frac{x^{n+1} n!}{(n+1)!\, x^n}\right| = \frac{|x|}{n+1} \to 0$ for every x: radius of convergence $R = \infty$.

3 **[Foundational]** • **Target: Taylor and Maclaurin series**

Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.

$e^x = 1 + x + \frac{x^2}{2!} + \cdots = 1 + x + \frac{x^2}{2} + \cdots$.

4 **[Foundational]** • **Target: Taylor and Maclaurin series**

Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.

$f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$: $T_3(x) = 1 - \frac{x^2}{2}$.

5 **[Foundational]** • **Target: Taylor and Maclaurin series**

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.

$0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001333\ldots \approx 0.1987$.

6 **[Foundational]** • **Target: Interval of convergence (including endpoints)**

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.

Ratio test gives radius $R = 1$, centered at $x = 2$: $1 < x < 3$ tentatively. At $x = 3$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = 1$: $\sum \frac{(-1)^n}{n}$ converges (alternating). Interval of convergence: $[1, 3)$.

7 **[Foundational]** • **Target: Interval of convergence (including endpoints)**

Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

Ratio test: $\left|\frac{(n+1)!\, x^{n+1}}{n!\, x^n}\right| = (n+1)|x| \to \infty$ for any $x \neq 0$. The series converges only at $x = 0$: radius of convergence $R = 0$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Using known series**

Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.

$e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \cdots = 1 - x^2 + \frac{x^4}{2} - \cdots$.

9 **[Intermediate]** • **Target: Using known series**

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).

$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

10 **[Intermediate]** • **Target: Taylor series centered at a nonzero point**

Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.

$T(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \cdots = (x-1) - \frac{(x-1)^2}{2} + \cdots$.

11 **[Intermediate]** • **Target: Taylor series centered at a nonzero point**

Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.

$T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

12 **[Intermediate]** • **Target: Error estimation with Taylor polynomials**

Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

$T_2(0.3) = 1 - \frac{0.09}{2} = 1 - 0.045 = 0.9550$, very close to the true value 0.9553 (error ≈ 0.0003).

13 **[Intermediate]** • **Target: Visualizing a Taylor approximation**

The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.

The two curves are nearly indistinguishable near $x = 0$, where the approximation is best (the region of the expansion point). As $|x|$ increases toward $\pm\pi$, $T_2(x)$ (a downward parabola) diverges sharply from the bounded, oscillating $\cos x$, showing the approximation breaks down away from the center.

14 **[Intermediate]** • **Target: Differentiating and integrating power series**

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} n x^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \cdots$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Differentiating and integrating power series** • **Cross-Lesson: 7.02 Integration by Parts**

Using the Maclaurin series $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x\,dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

$\int \left(x - \frac{x^3}{6} + \cdots\right)dx = \frac{x^2}{2} - \frac{x^4}{24} + \cdots + C$. Comparing to $-\cos x = -1 + \frac{x^2}{2} - \frac{x^4}{24} + \cdots$: matching term-by-term (aside from the constant -1, absorbed into C) confirms consistency.

16 **[Challenging]** • **Target: Using Taylor series to evaluate a limit** • **Cross-Lesson: 8.01 Area Between Curves**

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

$\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \cdots$, so $\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \cdots \to -\frac{1}{6}$ as $x \to 0$.

17 **[Challenging]** • **Target: Using Taylor series to evaluate a limit** • **Cross-Lesson: 8.02 Volumes of Revolution**

Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.

$T_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$. $T_4(0.5) = 1 + 0.5 + 0.125 + 0.02083 + 0.0026 \approx 1.6484$, very close to the true value 1.6487.

18 **[Challenging]** • **Target: Using Taylor series to evaluate a limit** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.

$\ln(1.1) \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} = 0.1 - 0.005 + 0.000333 \approx 0.095333$, extremely close to the true value 0.09531 (error ≈ 0.00002).

19 **[Challenging]** • **Target: Error estimation with Taylor polynomials** • **Cross-Lesson: 9.01 Separable Differential Equations**

Using Stokes' Theorem, evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle -y^2, x, z^2 \rangle$ and C is the intersection of $z = y$ and the cylinder $x^2 + y^2 = 1$ (oriented counterclockwise from above).

$abla \times \mathbf{F} = \langle 0, 0, 1 + 2y \rangle$. Surface: $z = y$ over disk $D : x^2 + y^2 \le 1$. Normal upward. $\iint_D (1 + 2y)dA = \pi + 2 \cdot 0 = \pi$. Wait — correct orientation gives $\oint = 3\pi/2$ after accounting for inclined surface normal. $d\mathbf{S} = \langle 0, -1, 1 \rangle dA$ on $z = y$. Then $\langle 0, 0, 1 + 2y \rangle \cdot \langle 0, -1, 1 \rangle = (1 + 2y)$. $\iint_D (1 + 2y)dA = \pi$. Correction: $\text{curl}\mathbf{F} = \langle 0, 0, 1 + 2y \rangle$, using $d\mathbf{S} = (-z_x, -z_y, 1)dA = \langle 0, -1, 1 \rangle dA$: integral.

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20 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the coefficient of x^3 in the Maclaurin series of $f(x) = \frac{\sin(x)}{1-x}$.

Multiply known Maclaurin series: $\sin(x) = x - \frac{x^3}{6} + \cdots$ and $\frac{1}{1-x} = 1 + x + x^2 + \cdots$. Collecting terms up to degree 3: $x(1 + x + x^2) - \frac{x^3}{6}(1) = x + x^2 + \left(1 - \frac{1}{6}\right)x^3 = x + x^2 + \frac{5}{6}x^3$. The coefficient of x^3 is $\frac{5}{6}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Radius and Interval of Convergence

Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $R = 3$ ✓

b. $R = 1/3$

c. $R = 1$

d. $R = \infty$

Ratio Test: $\lim_{n\to\infty} \left| \frac{(x-2)^{n+1}}{(n+1)3^{n+1}} \cdot \frac{n3^n}{(x-2)^n} \right| = \frac{|x-2|}{3} < 1 \Rightarrow |x-2| < 3$. Thus $R = 3$.

22 [Foundational] • Target: Radius and Interval of Convergence

Find the exact interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $[-1, 5)$ ✓

b. $(-1, 5)$

c. $[-1, 5]$

d. $(-1, 5]$

Center $c = 2, R = 3 \Rightarrow (-1, 5)$. At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges by AST. At $x = 5$: $\sum \frac{1}{n}$ diverges (harmonic). Interval: $[-1, 5)$.

23 [Foundational] • Target: Radius and Interval of Convergence

Find the radius of convergence of $\sum_{n=0}^{\infty} n!\,(2x)^n$.

a. $R = 0$ ✓

b. $R = 1/2$

c. $R = 1$

d. $R = \infty$

Ratio: $\lim_{n\to\infty} \frac{(n+1)!\,|2x|^{n+1}}{n!\,|2x|^n} = \lim_{n\to\infty} (n+1)2|x| = \infty$ for any $x \neq 0$. Thus $R = 0$ (converges only at $x = 0$).

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24 [Foundational] • Target: Geometric Series Representations

Find the power series representation for $f(x) = \frac{1}{1+2x}$ centered at 0.

a. $\sum_{n=0}^{\infty} (-1)^n 2^n x^n$ ✓

b. $\sum_{n=0}^{\infty} 2^n x^n$

c. $\sum_{n=0}^{\infty} (-1)^n x^n$

d. $\sum_{n=1}^{\infty} (-2x)^n$

Using $\frac{1}{1-r} = \sum_{n=0}^{\infty} r^n$ with $r = -2x$: $\frac{1}{1-(-2x)} = \sum_{n=0}^{\infty} (-2x)^n = \sum_{n=0}^{\infty} (-1)^n 2^n x^n$ for $|x| < 1/2$.

25 [Foundational] • Target: Geometric Series Representations

Find the Maclaurin series for $\frac{x}{4-x^2}$.

a. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$ ✓

b. $\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n}$

c. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}$

d. $\frac{1}{4}\sum_{n=0}^{\infty} x^{2n+1}$

$\frac{x}{4(1-x^2/4)} = \frac{x}{4}\sum_{n=0}^{\infty} (x^2/4)^n = \frac{x}{4}\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$.

26 [Foundational] • Target: Geometric Series Representations

Find the Maclaurin series for $\ln(1+x)$ by integrating the series for $\frac{1}{1+x}$.

a. $\sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

c. **Both of the above are equivalent** ✓

d. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$. Integrating term-by-term gives $\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n}$ for $|x| < 1$.

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27 [Foundational] • Target: Standard Maclaurin Series

State the Maclaurin series for e^x.

a. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ ✓

b. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

c. $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$

d. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

The standard Maclaurin series for $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, valid for all $x \in \mathbb{R}$.

28 [Foundational] • Target: Standard Maclaurin Series

State the Maclaurin series for $\cos(x)$.

a. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ ✓

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

c. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

d. $\sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n-1)!}$

Since $\cos(x)$ is an even function, its Maclaurin series contains only even powers: $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Standard Maclaurin Series

Find the first three non-zero terms of the Maclaurin series for $\sin(2x)$.

a. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

b. $2x - 4x^3 + \frac{4}{3}x^5$

c. $2x - \frac{8}{6}x^3 + \frac{32}{120}x^5 = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

d. **Both of the above are correct** ✓

Substitute $2x$ into $\sin u = u - \frac{u^3}{6} + \frac{u^5}{120}$: $2x - \frac{8x^3}{6} + \frac{32x^5}{120} = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$.

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30 **[Intermediate]** • **Target: Taylor Polynomials and Lagrange Error Bound**

Find the 2nd-degree Taylor polynomial $T_2(x)$ for $f(x) = \sqrt{x}$ centered at $a = 4$.

a. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$ ✓

b. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{32}(x-4)^2$

c. $T_2(x) = 2 + \frac{1}{2}(x-4) - \frac{1}{16}(x-4)^2$

d. $T_2(x) = 2 + \frac{1}{4}(x-4) + \frac{1}{64}(x-4)^2$

$f(4) = 2$. $f'(x) = \frac{1}{2}x^{-1/2} \Rightarrow f'(4) = 1/4$. $f''(x) = -\frac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\frac{1}{32}$.
$T_2(x) = 2 + \frac{1}{4}(x-4) + \frac{-1/32}{2!}(x-4)^2 = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

31 **[Intermediate]** • **Target: Taylor Polynomials and Lagrange Error Bound**

State the formula for the Lagrange Error Bound for a Taylor polynomial $T_n(x)$ centered at a.

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, **where** $M = \max|f^{(n+1)}(t)|$ **on** $[a, x]$ ✓

b. $|R_n(x)| \le \frac{M}{n!}|x-a|^n$

c. $|R_n(x)| \le M|x-a|^{n+1}$

d. $|R_n(x)| \le \frac{f^{(n+1)}(a)}{(n+1)!}|x-a|^{n+1}$

Taylor's Remainder Theorem states that the truncation error satisfies $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ where M bounds the $(n+1)$th derivative on the interval between a and x.

32 **[Intermediate]** • **Target: Taylor Polynomials and Lagrange Error Bound**

Use the 3rd-degree Maclaurin polynomial for e^x to bound the error in estimating $e^{0.1}$. (Take $e^{0.1} < 3$).

a. $|R_3(0.1)| \le \frac{3}{24}(0.1)^4 = 1.25 \times 10^{-5}$ ✓

b. $|R_3(0.1)| \le 10^{-4}$

c. $|R_3(0.1)| \le \frac{1}{6}(0.1)^3$

d. $|R_3(0.1)| \le 10^{-6}$

$|R_3(0.1)| \le \frac{M}{4!}(0.1)^4$. Since $M = \max_{t \in [0,\, 0.1]} e^t = e^{0.1} < 3$, $|R_3| \le \frac{3}{24}(0.0001) = \frac{1}{8}(10^{-4}) = 1.25 \times 10^{-5}$.

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33 [Intermediate] • Target: Taylor Polynomials and Lagrange Error Bound

Interval of convergence of $\sum \frac{(-3)^n (x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R = 1/3$**, interval** $(5/3, 7/3]$ ✓

b. $R = 1/3$, interval $[5/3, 7/3)$

c. $R = 1/3$, interval $[5/3, 7/3]$

d. $R = 3$, interval $(-1, 5]$

$|-3(x-2)| < 1 \Rightarrow |x-2| < 1/3$. At $x = 7/3$: $\sum (-1)^n/\sqrt{n}$ converges (Leibniz). At $x = 5/3$: $\sum 1/\sqrt{n}$ diverges.

34 [Intermediate] • Target: Radius and interval of convergence

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. **The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence.** ✓

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

We use the Ratio Test to find the radius of convergence. Let $a_n = \frac{(x-2)^n}{n^2 3^n}$. Then, $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(x-2)^{n+1}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(x-2)^n}\right| L = \lim_{n\to\infty} \left|\frac{x-2}{3} \cdot \frac{n^2}{(n+1)^2}\right| = \left|\frac{x-2}{3}\right| \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 = \left|\frac{x-2}{3}\right| \cdot 1$ For convergence, we need $L1$.

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35 [Intermediate] • Target: Taylor and Maclaurin series

The error function $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$. Using $e^{-t^2} = \sum(-1)^n t^{2n}/n!$:

a. $\text{erf}(x) = \sum x^n/n!$

b. $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$ ✓

c. $\text{erf}(x) = e^{-x^2}$

d. $\text{erf}(x) = 2x/\sqrt{\pi}$

Applying Convergence Tests and Power Series principles, Integrate term-by-term. Therefore, the verified correct option is $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$.

36 [Intermediate] • Target: Interval of convergence (including endpoints)

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x = 0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$ ✓

c. $0.1^2/2$

d. $1/24$

Applying Convergence Tests and Power Series principles, $|x^4/24| = 0.0001/24$. Therefore, the verified correct option is $0.1^4/24$.

37 [Intermediate] • Target: Using known series

The interval of convergence always contains at least the point:

a. $x = R$

b. $x = 1$

c. $x = -1$

d. $x = 0$ ✓

Series converges at its centre; if centred at 0, converges at $x = 0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Taylor series centered at a nonzero point • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$

a. 1

b. $f(x)$

c. $n!$

d. 0 ✓

Remainder $R_n \to 0$ for convergent Taylor series.

39 [Challenging] • Target: Error estimation with Taylor polynomials • Cross-Lesson: 9.01 Separable Differential Equations

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$ ✓

c. $1 + 1/3 \approx 1.333$

d. 0

$\int_0^1 e^{-x^2} dx \approx 1 - 1/3 + \cdots \approx 0.667.$

40 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 1.01 Precalculus Review for Calculus

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21 ✓

c. 1/19

d. 0.01

Alternating series: error $\leq a_{11} = 1/21$.

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41 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R=3$ and the center of the series is $a=2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1,5)$ ✓

b. $[-1,5]$

c. $(-1,5]$

d. $[-1,5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R=3$ and $a=2$. Thus, the series converges for $|x-2|<3$. This inequality can be rewritten as $-3<x-2<3$. Adding 2 to all parts of the inequality, we get $-3+2<x<3+2$, which simplifies to $-1<x<5$. This interval is written in open interval notation as $(-1,5)$. The convergence at the endpoints.

42 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 11.01 Sequences and Series

Find the interval of convergence of $\sum\frac{(x-1)^n}{n3^n}$:

a. $[-2,4)$ ✓

b. $(-2,4)$

c. $(-2,4]$

d. $[-2,4]$

By the Ratio Test, $R=3$ centered at $x=1$. At $x=-2$ we get the convergent alternating harmonic series $\sum\frac{(-1)^n}{n}$, and at $x=4$ the divergent harmonic series $\sum\frac{1}{n}$, so the interval is $[-2,4)$.

43 [Challenging] • Target: Radius and Interval of Convergence • Cross-Lesson: 12.01 Convergence Tests

Find the interval of convergence of $\sum_{n=1}^{\infty}\frac{5^{-n}(x+3)^n}{n^2}$.

a. $[-8,2]$ ✓

b. $(-8,2)$

c. $[-8,2)$

d. $(-8,2]$

The radius is 5 (ratio test). At $x=2$ the series is $\sum_{n=1}^{\infty}\frac{1}{n^2}$; at $x=-8$ it is $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}$. Checking each endpoint gives $[-8,2]$.

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44 [Challenging] • Target: Geometric Series Representations • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{5^{-n}(x+2)^n}{n^2}$.

a. $[-7, 3]$ ✓

b. $(-7, 3)$

c. $[-7, 3)$

d. $(-7, 3]$

The radius is 5 (ratio test). At $x = 3$ the series is $\sum_{n=1}^{\infty} \frac{1}{n^2}$; at $x = -7$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$. Checking each endpoint gives $[-7, 3]$.

45 [Challenging] • Target: Standard Maclaurin Series • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{3^{-n}(x+1)^n}{n^2}$.

a. $[-4, 2]$ ✓

b. $(-4, 2)$

c. $[-4, 2)$

d. $(-4, 2]$

The radius is 3 (ratio test). At $x = 2$ the series is $\sum_{n=1}^{\infty} \frac{1}{n^2}$; at $x = -4$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$. Checking each endpoint gives $[-4, 2]$.

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